Graphical solutions

1 Consider the function $y = \cos(x - 30)$.

- **a** Sketch the graph of the function for $-360 \leq x \leq 360$.
- **b** Sketch the line $y = \frac{1}{2}$ on the same number plane.
- **c** Hence, state all solutions to the equation $\cos(x - 30) = \frac{1}{2}$ over the domain $(-360, 360]$.

2 Consider the function $y = \sin(x - 60)$.

- **a** Sketch the graph of the function for $-360 \leq x \leq 360$..
- **b** Sketch the line $y = \frac{1}{2}$ on the same number plane.
- **c** Hence, state all solutions to the equation $\sin(x - 60) = \frac{1}{2}$ over the domain $[-360, 360)$.

3 Consider the function $y = 2\sin 4x$.

- **a** Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.
- **b** Sketch the line $y = 1$ on the same number plane.
- **c** Hence, state all solutions to the equation $2\sin 4x = 1$ over the domain $[-90^\circ, 90^\circ]$. Give your answers in degrees.

4 Consider the function $y = 2\sin 2x$.

- **a** Sketch the graph of the function for $-180^\circ \leq x \leq 180^\circ$.
- **b** State the other function you would add to the graph in order to solve the equation $2\sin 2x = 1$.
- **c** Sketch the graph of this function on the same number plane.
- **d** Hence, state all solutions to the equation $2\sin 2x = 1$ over the domain $[-180^\circ, 180^\circ]$.

5 Consider the function $y = 3\cos 2x + 1$.

- **a** Sketch the graph of the function for $-180 \leq x \leq 180$.
- **b** State the other function you would add to the graph in order to solve the equation $3\cos 2x + 1 = \frac{5}{2}$.
- **c** Sketch the graph of this function on the same number plane.
- **d** Hence, state all solutions to the equation $3\cos 2x + 1 = \frac{5}{2}$ over the domain $[-180, 180]$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Consider the function $y = 2\sin 3x - 3$.

- **a** Sketch the graph of the function for $-60 \leq x \leq 60$.
- **b** State the other function you would add to the graph in order to solve the equation $2\sin 3x - 3 = -2$.
- **c** Sketch the graph of this fuction on the same number plane.
- **d** Hence, state all solutions to the equation $2\sin 3x - 3 = -2$ over the domain $[-120, 120]$.

7 Consider the function $y = -2\cos 3x$.

- **a** Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.
- **b** State the other function you would add to the graph in order to solve the equation $-2\cos 3x = -1$.
- **c** Sketch the graph of this function on the same number plane.
- **d** Hence, state all solutions to the equation $-2\cos 3x = -1$ over the domain $[-120^\circ, 120^\circ]$.

Technology

8 Solve each equation for the given interval. Round your answers to one decimal place.

- **a** $\cos 2x = 0.9$ for $0 \leq x < 360$
- **b** $3\sin 2x = 1.2$ for $0 \leq x < 180$
- **c** $4\cos\left(\frac{x}{3} + 2\right) = 1$ for $0 \leq x < 720$

9 Consider the equation $\sin\left(\frac{x}{2} + 60^\circ\right) = \cos\left(\frac{x}{2} - 60^\circ\right)$.

- **a** State the two functions you would graph in order to solve this equation graphically.
- **b** Sketch the graph of both of these functions using technology.
- **c** Hence, state all solutions to the equation over the domain $[-360^\circ, 360^\circ]$.

10 Use technology to solve the following functions over the interval $[0^\circ, 360^\circ)$. Give all solutions to the nearest degree.

- **a** $2\sin 3x - 4\sin 2x = 0$
- **b** $2\sin x + 5\cos x = 1$
- **c** $\cos 3x + \cos x = \sin x$
- **d** $2\cos 3x - 3\cos 2x = 0$
- **e** $\cos\left(\frac{x}{2}\right) - 4\sin 2x = 0$
- **f** $\sin\left(\frac{x}{2}\right) = 7\cos 3x$
- **g** $4\sin^5(x) = -(\cos x + 1)$
- **h** $\cos^3(x) + \cos x = -1$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Use technology to solve the following equations for $0 \leq x \leq 360$. Give all solutions to two decimal places.

a $2\sin x + 5\cos x = 1$

b $3\sin 2x + 2\sin 3x = 0$

c $3\sin(2x) = x + 1$

d $2\cos 3x + 3\cos 2x = 0$

e $\cos 3x + \cos x = \sin x$

f $\cos\left(\frac{x}{2}\right) - 4\sin 2x = 0$

g $4\sin^5(x) = -(\cos x + 1)$

h $\cos^5(x) + \cos x = -1$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647